1 · INGESTION

Upload a document
pdf · markdown/web · image · code

Parser router
document_cascade.py

only if flagged

Deterministic
Docling · Trafilatura

LLM / vision
only if flagged
(tables, OCR, order)

Chunking
384 tok/chunk, 64 overlap

Embed chunks
nomic-embed-text, local

write: parent + map

write: index chunk text (BM25)

write: child vectors

2 · QUERY UNDERSTANDING

Ask a question
+ conversation context

Query Analyzer
patterns first, else LLM decision

strategy?

direct / decompose / step-back / conversation

Formulate 1–2 queries
one retrieval pass per query, below

Embed query
dense vector

Lexical terms
BM25 tokens, raw text

read: kNN, top-20

read: BM25, top-20

3 · SHARED STORAGE

Document Store
whole document, SQLite
+ chunk → parent map

Vector Index
Chroma — child chunk embeddings
cosine similarity search

Lexical Index
SQLite FTS — BM25 over
child chunk text

4 · RETRIEVAL

Hybrid fusion
Reciprocal Rank Fusion, k=60

Cross-encoder rerank
ms-marco-MiniLM-L6-v2

re-run fusion for the rewritten need

read: fetch parent doc

Resolve parent documents
chunk hits → full parent context

5 · EVIDENCE & CORRECTION

Evidence judges
deterministic corroboration / anchor
shortcut first — LLM only for the middle band

evidence sufficient?

no

Rewrite the need
Ollama, one attempt

Corrective retrieval
re-enters Retrieval, above

yes

Context Builder
≤3 parents/need, ≤3000 tokens

6 · GENERATION

LLM Generation
Groq gpt-oss-120b
→ Ollama qwen3:14b fallback
told every information need to cover

attempts remain

7 · VERIFICATION

Verify claims
deterministic checks first —
structured, exact, lexical, embedding
LLM judge only on a miss

claims supported?

no

Regenerate
attempts remain, fed the gap

yes

Finalize
grounded, cited answer

no

no

Search the web
retries exhausted, web allowed

Abstain
exhausted, no web allowed

no reliable answer